

Lazarus - Management Plan

astroFUEL

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Madrid, October 2026

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Acronyms

AIT Assembly, Integration and Testing.

AOCS/GNC Attitude and Orbit Control System/Guidance, Navigation and Control.

CDR Critical Design Review.

ECSS European Cooperation for Space Standardization.

EOL End of Life.

EOM End of Mission.

FAR Flight Acceptance Review.

FEM Finite Element Method.

FSW Flight Software.

GEO Geostationary Orbit.

GTO Geosynchronous Transfer Orbit.

LEOP Launch and Early Orbit Phase.

MDR Mission Definition Review.

MLI Multi-layer insulation.

PBS Project Breakdown Structure.

PDR Preliminary Design Review.

PRR Preliminary Requirements Review.

QR Qualification Request.

RVM Rendezvous Mode.

SFM Safe Mode.

SRR System Requirements Review.

WBS Work Breakdown Structure.

WP Work Package.

1 Introduction

Purpose of the document, mission overview and scope.

Usamos Work Package (WP) Rendezvous Mode (RVM) Safe Mode (SFM) Geostationary Orbit (GEO), Geosynchronous Transfer Orbit (GTO), Attitude and Orbit Control System/Guidance, Navigation and Control (AOCS/GNC)

2 Project Organization

2.1 Identification of the customer

Customer, stakeholders and the management requirements they impose.

2.2 Company Organization

AstroFUEL is a private initiative in the European new space market. The company manages the complete spacecraft development process in-house, from mission analysis and subsystem design through manufacturing, integration, and final assembly. This approach provides full control over the spacecraft development process and allows the company to make decisions independently, while also enabling it to respond rapidly to new business opportunities arising in the constantly evolving new space market.

External suppliers are used exclusively for the procurement of the materials, components, and parts required for the spacecraft. The integration and assembly of these elements into the spacecraft are carried out entirely in-house by astroFUEL.

To enable fast and effective decision-making, the company follows a clear chain of command. The CEO is the head of the company and oversees all ongoing projects, allocating the available company resources to each project according to its needs. The CEO appoints a Project Manager for each project, who is responsible for initiating, planning, executing, monitoriong, controlling and closing the project. This includes scope, time, cost, quality, risk and human resource management.

Under the responsibility of the Project Manager, the technical development of each project is divided into different areas. Since astroFUEL's first project, LAZARUS, is for now only a technology demonstrator at a preliminary design stage, the company has only technical areas, pending the move to operate as a commercial enterprise. That is when the legal, HR, commercial, and environmental departments, among others, will be established.

The technical development of LAZARUS is divided into the following areas: Propulsion, Mission Analysis, Mechanical & Structural Engineering, Communications & Data Handling, Thermal Management, Payload, Attitude & Orbit Control/Guidance, Navigation & Control, and Power Management (see Figure 2.1). Each technical area is responsible for the analysis, design, and development of its corresponding subsystem or technical domain, as well as for contributing to the integration of the complete spacecraft.

Mission Analysis is responsible for defining and analysing the mission scenarios and orbital requirements. Propulsion is responsible for the design and development of the propulsion system required for spacecraft manoeuvring. Attitude & Orbital Control/Guidance, Navigation & Control (AOCS/GNC) is responsible for the attitude and orbit control functions required throughout the

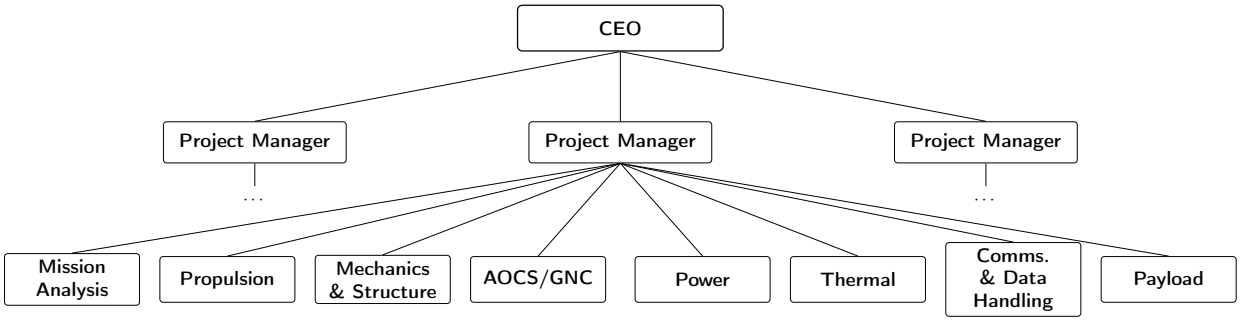


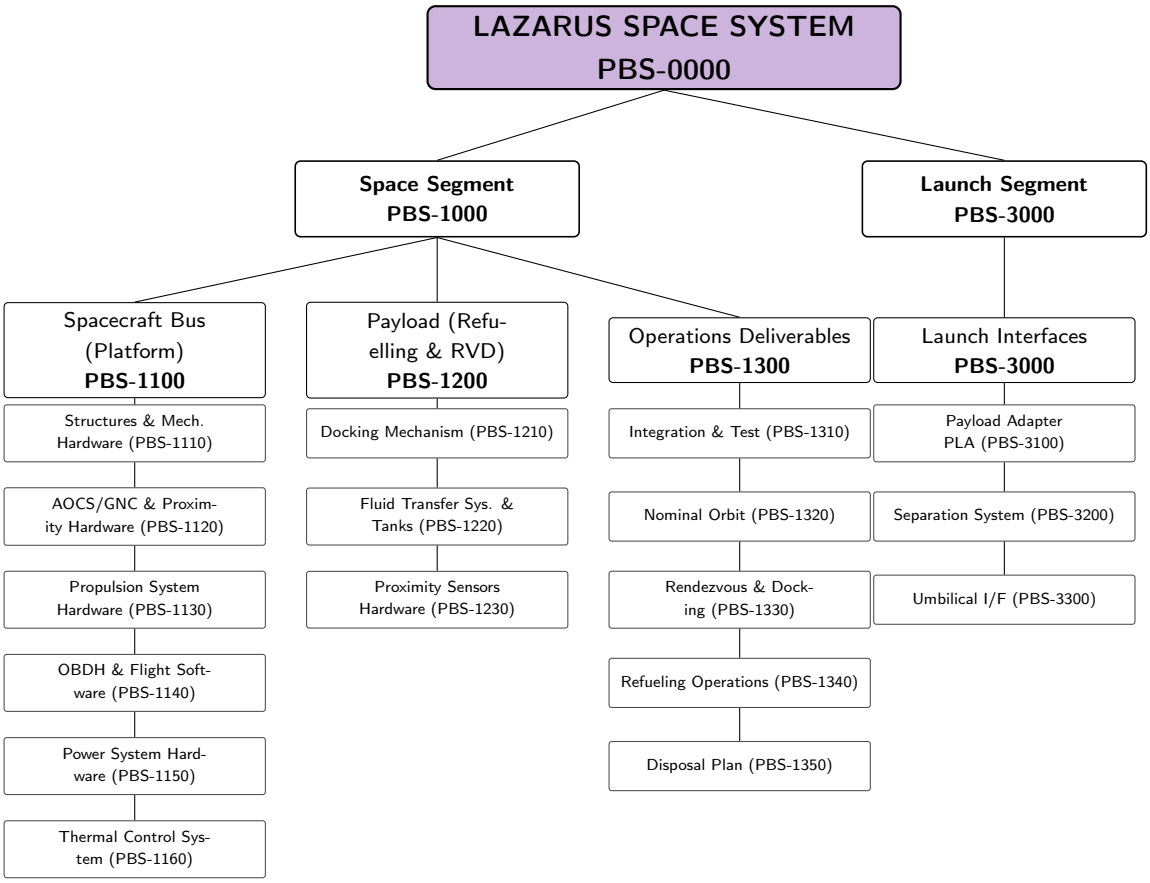
Figure 2.1: Schematic representation of the astroFUEL's internal company organization.

mission, including maintaining the target orbits and facing and performing the operations required to approach and service the client satellite. Mechanical & Structural Engineering is responsible for the mechanical design of the spacecraft, including the primary and secondary structures, as well as for the integration of all subsystems into the spacecraft structure. Power Management is responsible for the generation, storage, and distribution of electrical power throughout the spacecraft, ensuring that all subsystems can operate nominally throughout the mission. Thermal Management is responsible for the thermal design and analysis of the spacecraft, ensuring that all components and subsystems remain within their specified operating temperature ranges. Particular attention is given to the payload, which requires specific thermal conditions throughout the mission. Communications & Data Handling is responsible for the spacecraft's communication functions, onboard data handling, and data transmission. Finally, the Payload area is responsible for coordinating with the other technical areas the design and development of the equipment required to store and handle Xenon under the specified conditions throughout the mission, as well as to perform the servicing and refuelling of client satellites.

3 PBS

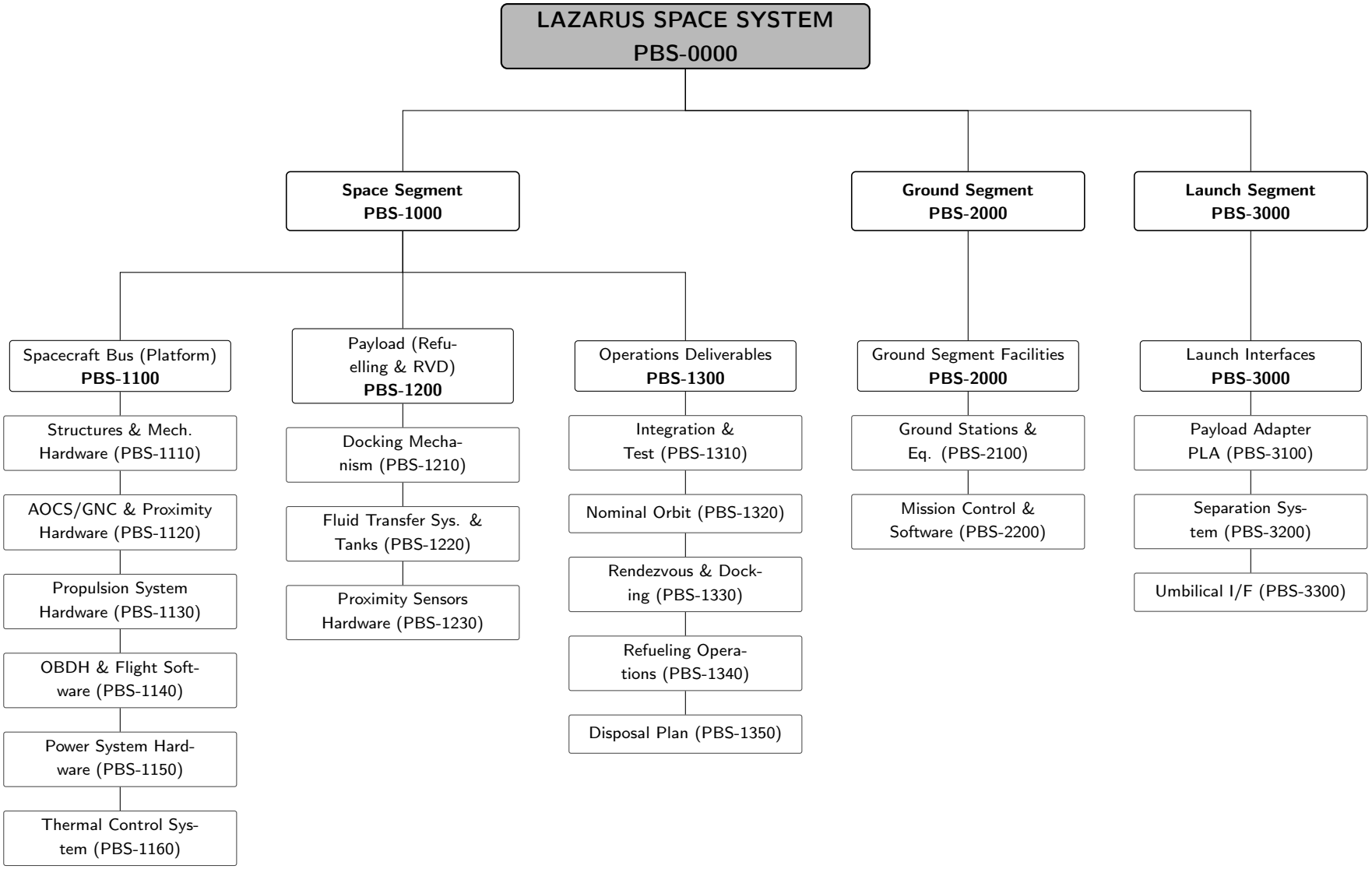
3.1 PBS of the Preliminary Design Project

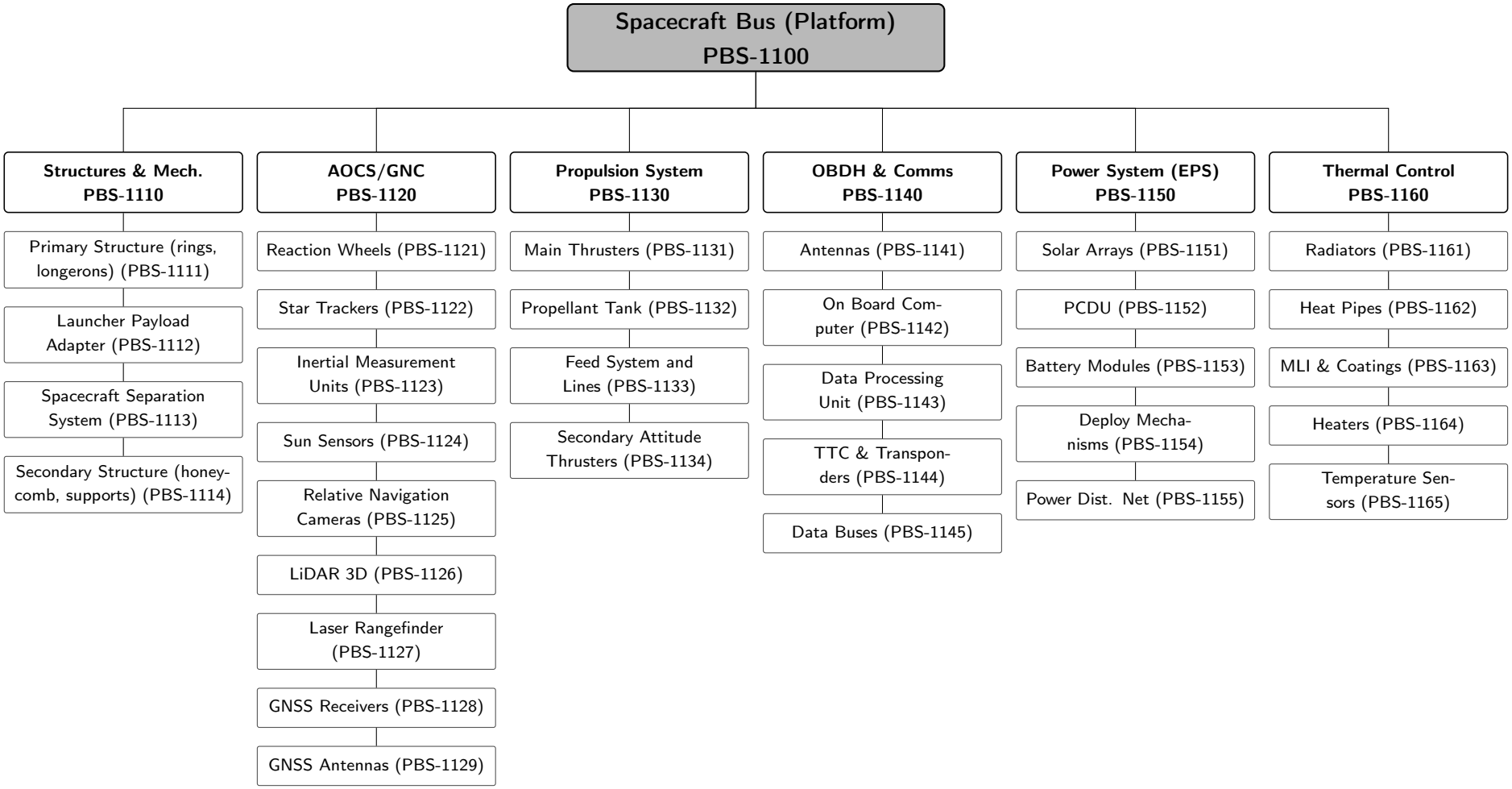
The Preliminary Design PBS outlines the physical and document deliverables required during the initial study and design phases (Phases 0, A, and B). At this stage, the structure focuses strictly on defining the core mission architecture: the Spacecraft Bus, the Refueling Payload, preliminary Operations Deliverables, and early Launch Interfaces. The Ground Segment is deliberately excluded from this preliminary tree, as its detailed development, validation, and procurement are scheduled to begin later in Phase C, ensuring strict alignment with the project's WBS and phasing.

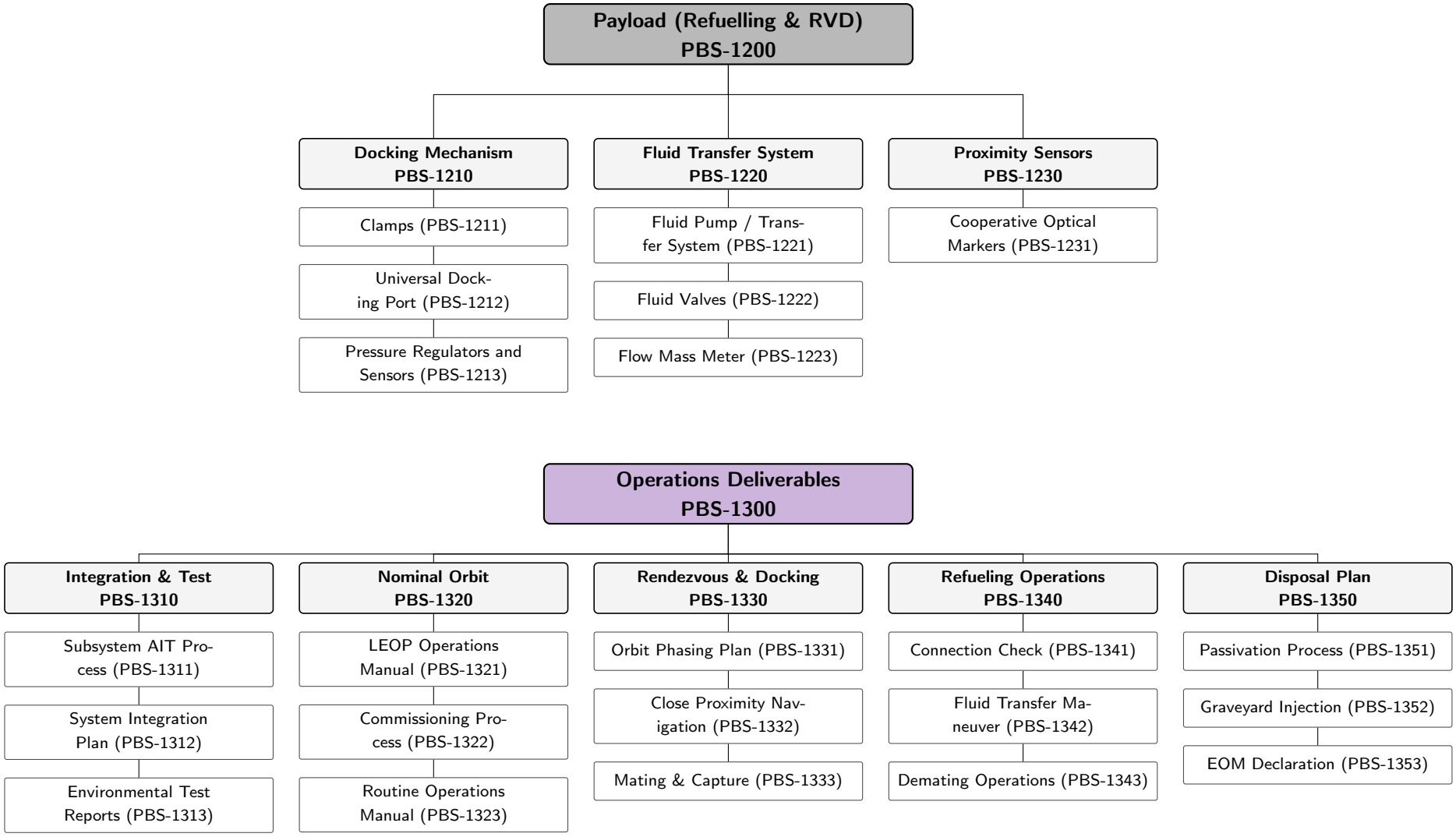


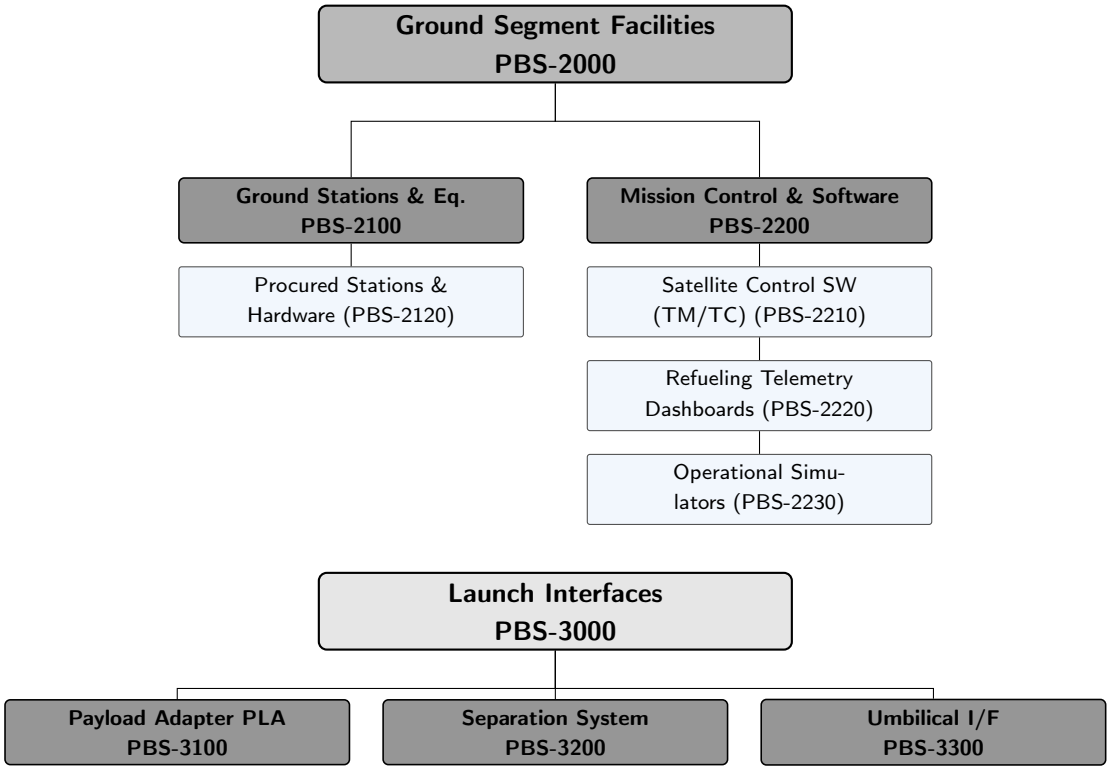
3.2 PBS of the complete Project

The Complete Project expands upon the preliminary foundation to represent the entire LAZARUS mission lifecycle, extending through End of Life (EOL) disposal (Phase F). It integrates the Ground Segment Facilities, including Ground Stations and Mission Control Software, and provides a comprehensive, lower-level breakdown of all major segments.


LAZARUS SPACE SYSTEM
PBS-0000
Space Segment
PBS-1000
Ground Segment
PBS-2000
Launch Segment
PBS-3000
Spacecraft Bus (Platform)
PBS-1100
Structures & Mech. Hardware (PBS-1110)
AOCS/GNC & Proximity Hardware (PBS-1120)
Propulsion System Hardware (PBS-1130)
OBDH & Flight Software (PBS-1140)
Power System Hardware (PBS-1150)
Thermal Control System (PBS-1160)
Payload (Refuelling & RVD)
PBS-1200
Docking Mechanism (PBS-1210)
Fluid Transfer Sys. & Tanks (PBS-1220)
Proximity Sensors Hardware (PBS-1230)
Operations Deliverables
PBS-1300
Integration & Test (PBS-1310)
Nominal Orbit (PBS-1320)
Rendezvous & Docking (PBS-1330)
Refueling Operations (PBS-1340)
Disposal Plan (PBS-1350)
Ground Segment Facilities
PBS-2000
Ground Stations & Eq. (PBS-2100)
Mission Control & Software (PBS-2200)
Launch Interfaces
PBS-3000
Payload Adapter PLA (PBS-3100)
Separation System (PBS-3200)
Umbilical I/F (PBS-3300)


Spacecraft Bus (Platform)
PBS-1100
Structures & Mech.
PBS-1110
Primary Structure (rings, longerons) (PBS-1111)
Launcher Payload Adapter (PBS-1112)
Spacecraft Separation System (PBS-1113)
Secondary Structure (honeycomb, supports) (PBS-1114)
AOCS/GNC
PBS-1120
Reaction Wheels (PBS-1121)
Star Trackers (PBS-1122)
Inertial Measurement Units (PBS-1123)
Sun Sensors (PBS-1124)
Relative Navigation Cameras (PBS-1125)
LiDAR 3D (PBS-1126)
Laser Rangefinder (PBS-1127)
GNSS Receivers (PBS-1128)
GNSS Antennas (PBS-1129)
Propulsion System
PBS-1130
Main Thrusters (PBS-1131)
Propellant Tank (PBS-1132)
Feed System and Lines (PBS-1133)
Secondary Attitude Thrusters (PBS-1134)
OBDH & Comms
PBS-1140
Antennas (PBS-1141)
On Board Computer (PBS-1142)
Data Processing Unit (PBS-1143)
TTC & Transponders (PBS-1144)
Data Buses (PBS-1145)
Power System (EPS)
PBS-1150
Solar Arrays (PBS-1151)
PCDU (PBS-1152)
Battery Modules (PBS-1153)
Deploy Mechanisms (PBS-1154)
Power Dist. Net (PBS-1155)
Thermal Control
PBS-1160
Radiators (PBS-1161)
Heat Pipes (PBS-1162)
MLI & Coatings (PBS-1163)
Heaters (PBS-1164)
Temperature Sensors (PBS-1165)


Payload (Refuelling & RVD)
PBS-1200
Docking Mechanism
PBS-1210
Clamps (PBS-1211)
Universal Docking Port (PBS-1212)
Pressure Regulators and Sensors (PBS-1213)
Fluid Transfer System
PBS-1220
Fluid Pump / Transfer System (PBS-1221)
Fluid Valves (PBS-1222)
Flow Mass Meter (PBS-1223)
Proximity Sensors
PBS-1230
Cooperative Optical Markers (PBS-1231)
Operations Deliverables
PBS-1300
Integration & Test
PBS-1310
Subsystem AIT Process (PBS-1311)
System Integration Plan (PBS-1312)
Environmental Test Reports (PBS-1313)
Nominal Orbit
PBS-1320
LEOP Operations Manual (PBS-1321)
Commissioning Process (PBS-1322)
Routine Operations Manual (PBS-1323)
Rendezvous & Docking
PBS-1330
Orbit Phasing Plan (PBS-1331)
Close Proximity Navigation (PBS-1332)
Mating & Capture (PBS-1333)
Refueling Operations
PBS-1340
Connection Check (PBS-1341)
Fluid Transfer Maneuver (PBS-1342)
Demating Operations (PBS-1343)
Disposal Plan
PBS-1350
Passivation Process (PBS-1351)
Graveyard Injection (PBS-1352)
EOM Declaration (PBS-1353)


Ground Segment Facilities
PBS-2000
Ground Stations & Eq.
PBS-2100
Mission Control & Software
PBS-2200
Procured Stations &
Hardware (PBS-2120)
Satellite Control SW
(TM/TC) (PBS-2210)
Refueling Telemetry
Dashboards (PBS-2220)
Operational Simu-
lators (PBS-2230)
Launch Interfaces
PBS-3000
Payload Adapter PLA
PBS-3100
Separation System
PBS-3200
Umbilical I/F
PBS-3300

4 Project Phasing

LAZARUS is divided into a series of sequenced and consecutive phases covering the entire mission's lifecycle, from the initial definition of the mission to the end of the spacecraft's operational lifetime. As outlined in Figure 4.1, each phase is composed of a number of tasks (white boxes) and milestones (purple boxes).

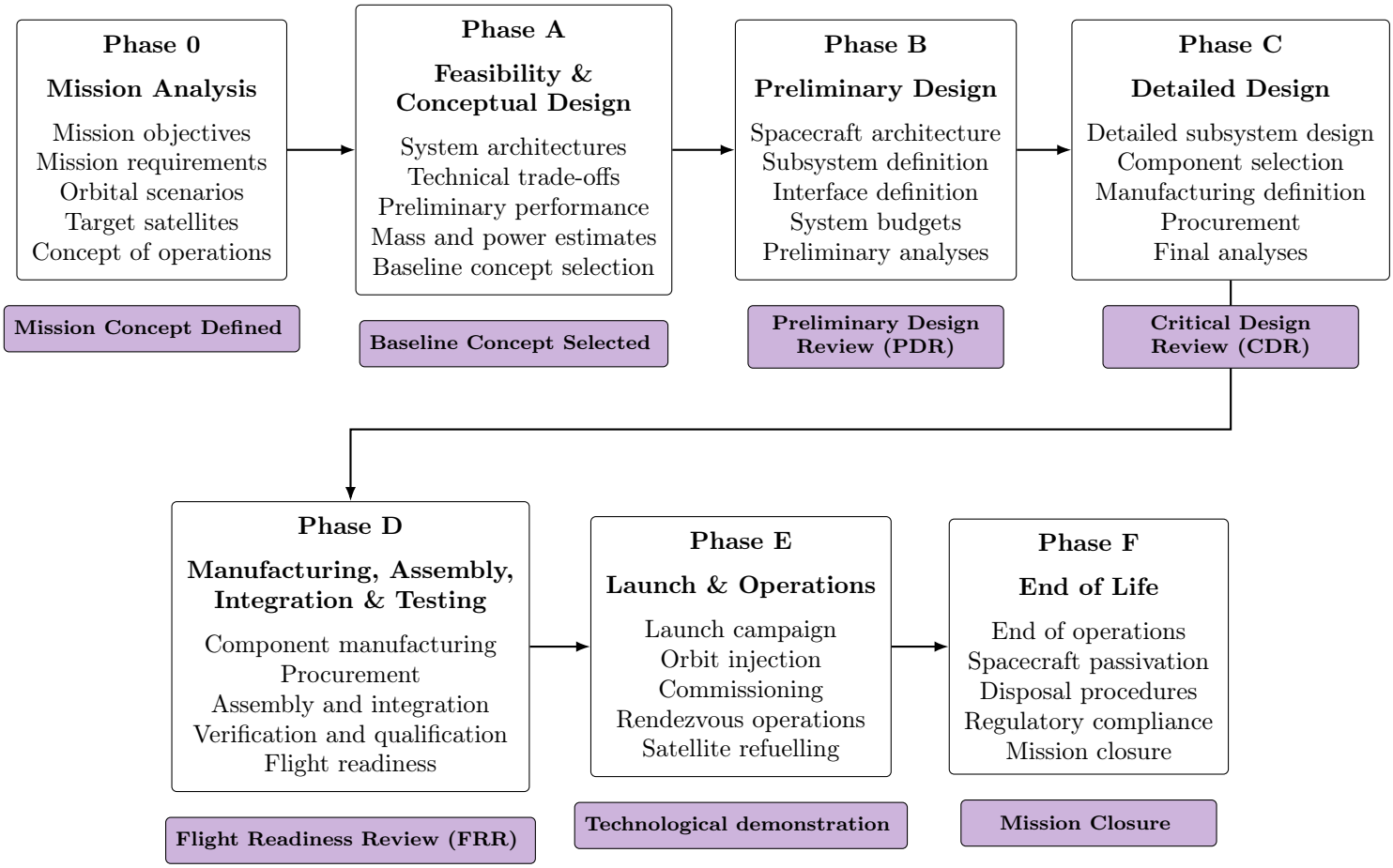


Figure 4.1: LAZARUS project phases, main activities and milestones.

The main project phases are defined as follows:

Phase 0 – Mission Analysis

During this phase, the mission objectives and concept of operations are defined. The main mission requirements, orbital scenarios, target satellites, and overall concept for the in-orbit refuelling mission are established. Initial feasibility assessments and system-level trade-offs are performed to identify the most suitable mission approach.

Phase A – Feasibility and Conceptual Design

The feasibility of LAZARUS is assessed during this phase. Different system architectures and technical solutions are analysed and compared. Preliminary estimates of spacecraft performance, mass, power, propulsion requirements, and other key system parameters are established. The phase concludes with the selection of the baseline mission concept.

Phase B – Preliminary Design

The selected mission concept is developed into a preliminary spacecraft design. The overall spacecraft architecture is defined, and the main subsystems are designed at a preliminary level. Interfaces between subsystems are established, and the main system budgets are defined and consolidated.

Phase C – Detailed Design

The preliminary design is further developed into a detailed design suitable for manufacturing, assembly, integration, and testing. Detailed analyses are performed for all spacecraft subsystems, and the required components and materials are specified. Components that cannot be manufactured in-house are procured from external suppliers.

Phase D – Manufacturing, Assembly, Integration and Testing

The spacecraft components are manufactured in-house or procured from external suppliers. The components are subsequently assembled and integrated by astroFUEL. The integrated spacecraft is subjected to the required verification and qualification tests to demonstrate compliance with the applicable system and subsystem requirements and to verify its readiness for launch.

Phase E – Launch and Operations

The spacecraft is launched and commissioned in orbit. Following successful commissioning, astroFUEL conducts the planned mission operations, including spacecraft control, rendezvous and proximity operations, and the refuelling of client satellites (only as a technology demonstrator).

Phase F – End of Life

At the end of its operational lifetime, the spacecraft is removed from active service and the appropriate EOL and disposal procedures are implemented in accordance with the applicable mission requirements and regulatory constraints.

5 Project planning

5.1 Project planning of the Preliminary Design Project

This section details the strategic and operational planning for the preliminary design of the Lazarus mission, covering the activities corresponding to Phases 0, A, and B. To ensure efficient execution, resource management has been structured based on the work packages previously defined in the Work Breakdown Structure (WBS).The Responsibility Assignment Matrix (RAM) documents the distribution of tasks among the technical areas using a standardized nomenclature. The R (Responsible) role identifies the main department in charge of leading. Meanwhile, the S (Support) role designates the areas acting as support.

Furthermore, this section establishes the logical diagram and the project's schedule. These schemas sequence the parallel development of the various subsystems through to their integration, culminating in the primary milestone that marks the conclusion of this stage: the Preliminary Design Review (PDR). This review will consolidate the satellite's baseline architecture and establish the critical path that will define the preliminary project.

Work Package	**PM**	**MA**	**M&S**	**Prop**	**AOCS/ GNC**	**Pwr**	**Thrm**	**C&D**	**Pay**
WP-1000	**R**								
WP-1100	**R**								
WP-1110	S		R						
WP-1110.1	S		R						S
WP-1110.2	S		R						
WP-1110.3	S		R						S
WP-1110.4	S		R	S					
WP-1110.5	S	S	R	S	S	S	S	S	S
WP-1110.6	S		R			S	S	S	S
WP-1120	S				R				
WP-1120.1	S	S			R			S	
WP-1120.2	S			S	R				
WP-1120.3	S	S			R			S	S
WP-1120.4	S		S		R	S			
WP-1130	S			R					
WP-1130.1	S	S		R					
WP-1130.2	S	S		R					
WP-1130.3	S		S	R					
WP-1130.4	S	S		R					
WP-1130.5	S		S	R	S				
WP-1130.6	S		S	R					S
WP-1140	S							R	
WP-1140.1	S	S						R	

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-1140.2	S				S			R	S
WP-1140.3	S							R	
WP-1140.4	S							R	
WP-1150	S					R			
WP-1150.1	S	S	S	S	S	R	S	S	S
WP-1150.2	S				S	R		S	
WP-1150.3	S		S			R			
WP-1150.4	S					R	S		
WP-1150.5	S					R		S	
WP-1160	S						R		
WP-1160.1	S	S	S	S	S	S	R	S	S
WP-1160.2	S	S					R		
WP-1160.3	S					S	R		
WP-1160.4	S		S				R		
WP-1200	**S**								**R**
WP-1210	S								R
WP-1210.1	S		S						R
WP-1220	S								R
WP-1220.1	S			S					R
WP-1220.2	S						S		R
WP-1230	S								R
WP-1230.1	S	S			S			S	R
WP-1300	**R**	**S**							
WP-1320	R	S							
WP-1320.1	R	S							
WP-1320.2	R	S		S					
WP-1330	R	S							
WP-1330.1	R	S			S				
WP-1340	R	S							
WP-1340.1	R	S			S				S
WP-1350	R	S							
WP-1350.1	R	S	S	S					
WP-3000	**R**	**S**	**S**						
WP-3100	R	S	S						
WP-3100.1	R	S	S						
WP-3100.2	R	S	S						
WP-3300	R	S							
WP-3300.1	R	S							

Table 5.1: Responsibility Assignment Matrix.

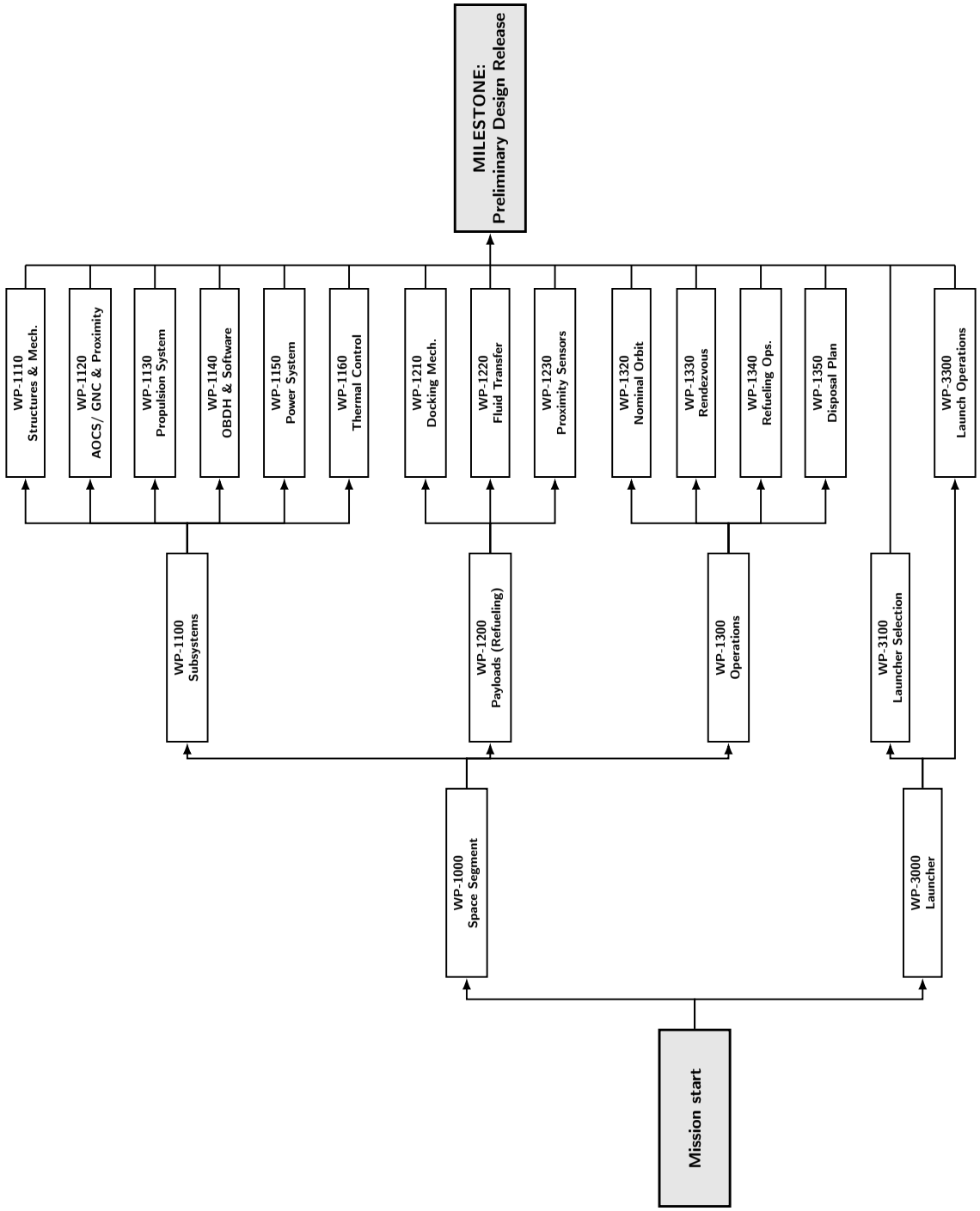


Figure 5.1: WBS logical diagram.

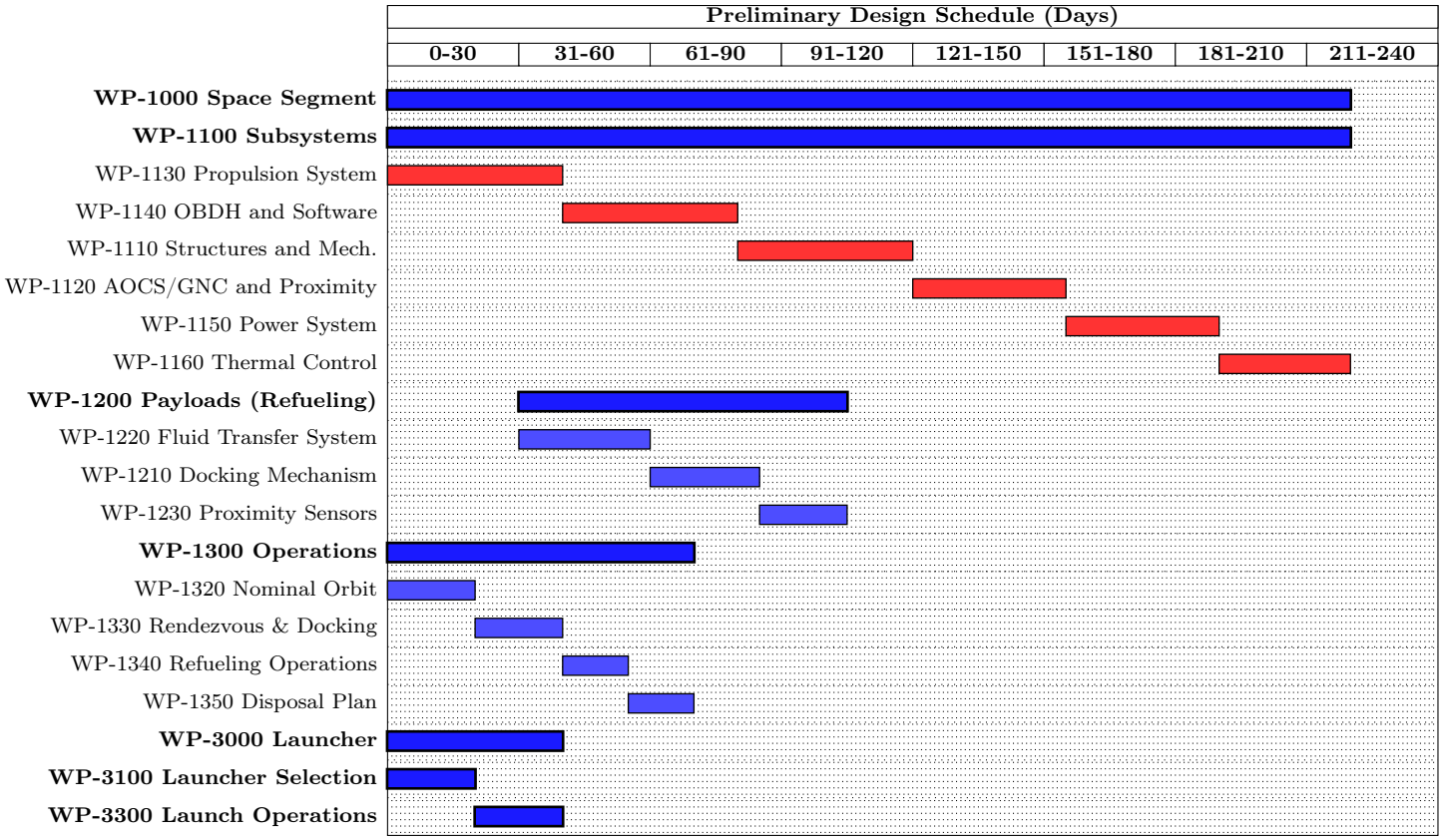


Figure 5.2: Preliminary Design Schedule (Phases 0, A, and B). The Critical Path through the subsystems is highlighted in red.

5.2 Project planning of the complete Project

Schedule of the complete project, based on the WBS of Section 7.2.

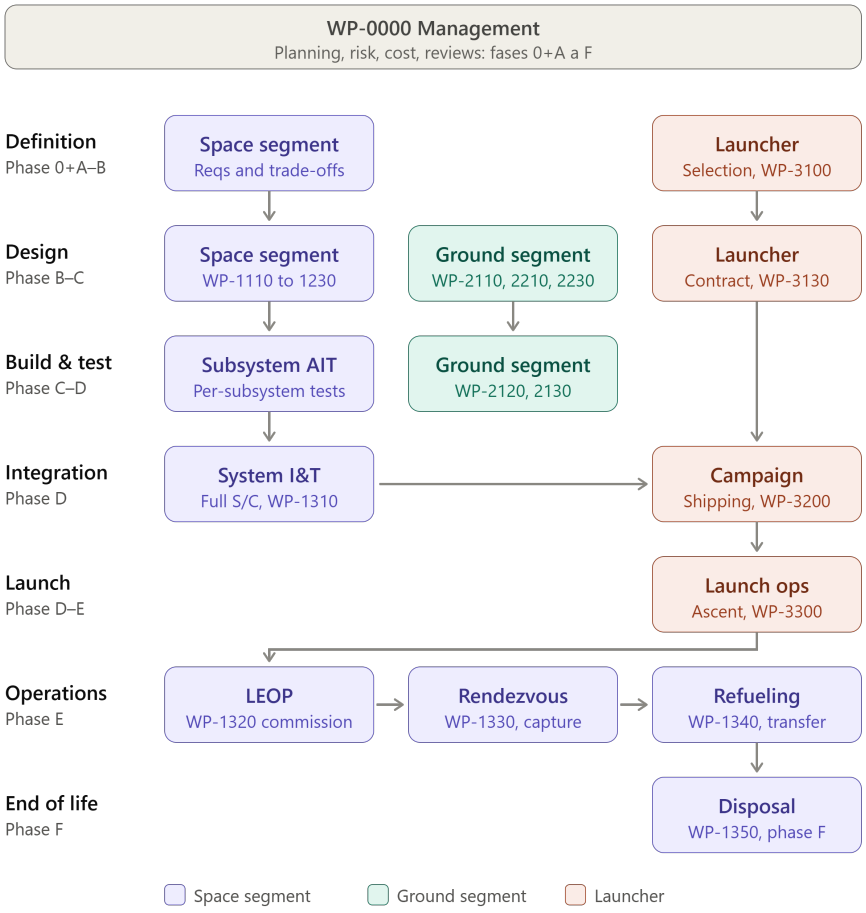


Figure 5.3: Logical flow of the complete project based on the WBS.

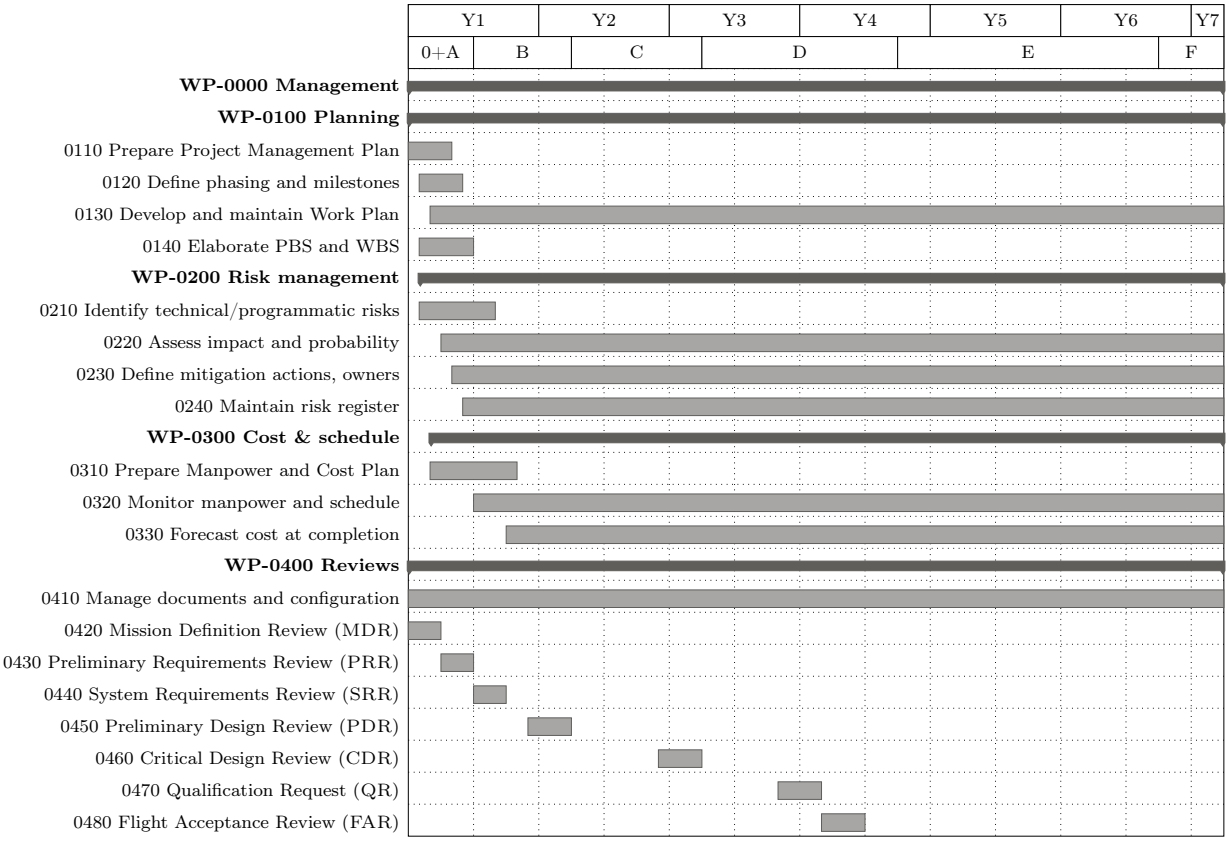


Figure 5.4: Schedule of WP-0000 Management (Level 4 tasks). Time is in project months.

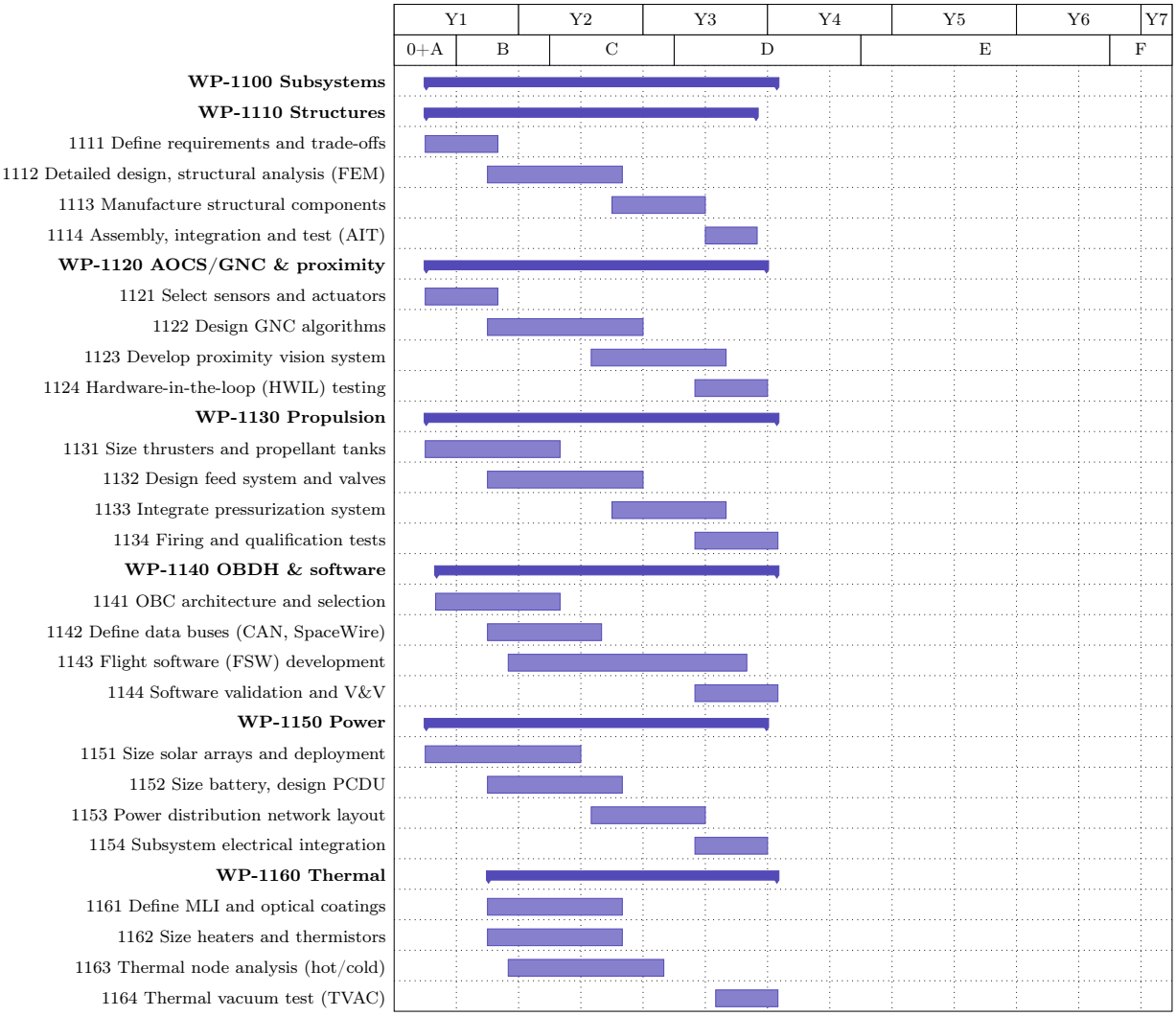


Figure 5.5: Schedule of WP-1100 Subsystems (Level 4 tasks). Time is in project months.



Figure 5.6: Schedule of WP-1200 Payloads (Refueling) and WP-1300 Operations (Level 4 tasks). Time is in project months.

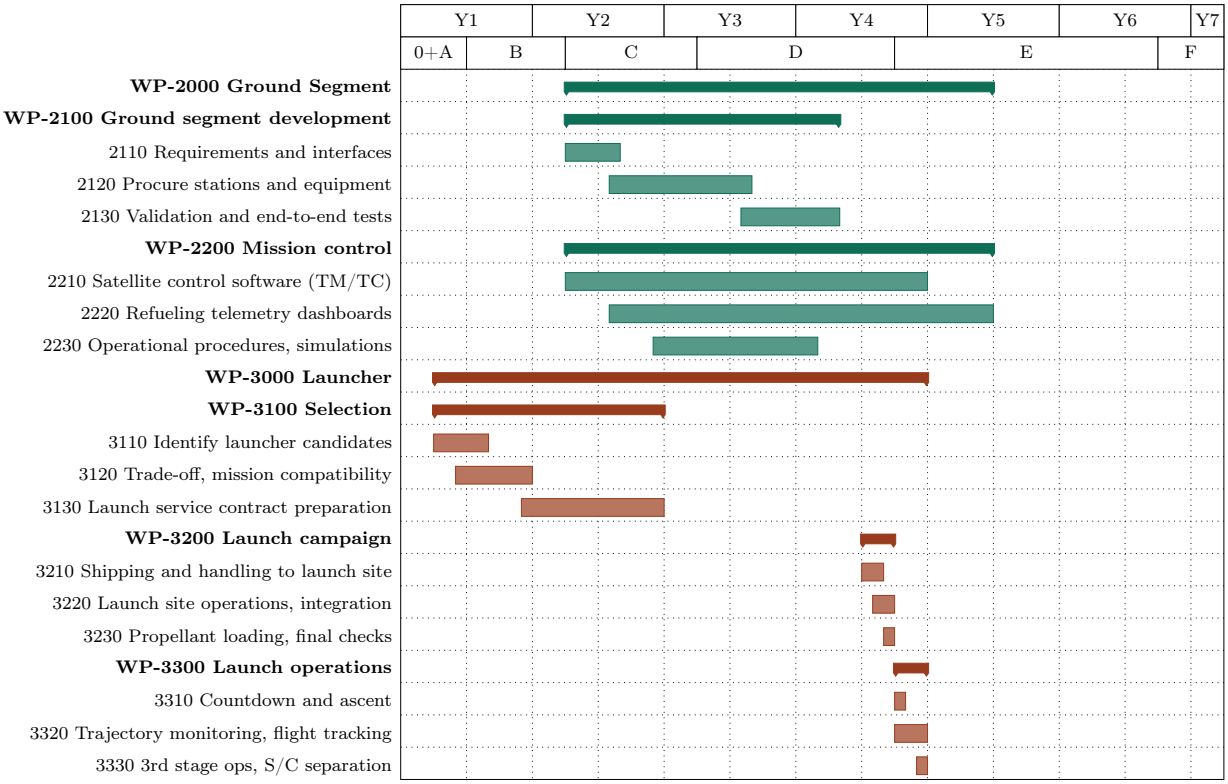


Figure 5.7: Schedule of WP-2000 Ground Segment and WP-3000 Launcher (Level 4 tasks). Time is in project months.

6 Risk Management

6.1 Identification of main risks

List of main risks.

6.2 Characterization of main risks in terms of impact and probability

Risk matrix.

7 WBS

7.1 WBS of the Preliminary Design Project

WBS limited to the work needed to reach the PDR (Phases 0/A and B). Same format as Section 7.2.

Table 7.1: Detailed scope of the LAZARUS WBS down to Level 4, including specific ECSS phases. Launch operations have been shifted to WP-3000.

WP	Scope	Phases
WP-0000 Management		
WP-0100 Planning	**0110:** Prepare the Project Management Plan.	0+A–B
	0120: Define the project phasing and milestones.	0+A–B
	0130: Develop and maintain the Project Work Plan.	0+A–F
	0140: Elaborate the PBS and WBS.	0+A–B
WP-0200 Risk Mgmt	**0210:** Identify main technical and programmatic risks.	0+A–B
	0220: Assess impact and probability (Risk Matrix).	0+A–F
	0230: Define mitigation actions and assign owners.	0+A–F
	0240: Maintain risk register and follow up.	0+A–F
WP-0300 Cost & Sched.	**0310:** Prepare Manpower and Cost Plan.	0+A–B
	0320: Monitor manpower and schedule vs baseline.	0+A–F
	0330: Forecast cost at completion and report.	0+A–F
WP-0400 Reviews	**0410:** Manage documents and configuration.	0+A–F
	0420: Prepare and hold MDR.	0+A
	0430: Prepare and hold PRR.	A
	0440: Prepare and hold SRR.	B
	0450: Prepare and hold PDR.	B
	0460: Prepare and hold CDR.	C
	0470: Prepare and hold QR.	D
	0480: Prepare and hold FAR.	D
WP-1000 Space Segment		
WP-1110 Structures	**1111:** Define requirements and trade-offs.	A–B
	1112: Detailed design and structural analysis, FEM.	B–C
	1113: Manufacturing of structural components.	C–D
	1114: AIT.	D
WP-1120 AOCS/GNC & Prox	**1121:** Selection of sensors and actuators.	A–B
	1122: Design of GNC algorithms.	B–C
	1123: Development of the proximity vision system.	C–D
	1124: Hardware-in-the-Loop (HWIL) testing.	D

Continued on next page

WP	Scope	Phases
WP-1130 Propulsion	**1131:** Thrusters and propellant tanks sizing.	A–C
	1132: Feed system and valves detailed design.	B–C
	1133: Pressurization system integration.	C–D
	1134: Subsystem firing and qualification tests.	D
WP-1140 OBDH & SW	**1141:** On-Board Computer architecture and selection.	A–C
	1142: Data buses definition (CAN, SpaceWire).	B–C
	1143: FSW development.	B–D
	1144: Software validation and V&V testing.	D
WP-1150 Power	**1151:** Solar arrays sizing and deployment mech.	A–C
	1152: Battery sizing and PCDU design.	B–C
	1153: Power distribution network layout.	C–D
	1154: Subsystem electrical integration.	D
WP-1160 Thermal	**1161:** Definition of MLI and optical coatings.	B–C
	1162: Sizing of heaters and thermistors.	B–C
	1163: Thermal node analysis (Hot/Cold cases).	B–C
	1164: Thermal Vacuum test (TVAC).	D
WP-1210 Docking Mech	**1211:** Mechanical design and interfaces.	A–C
	1212: Actuators, capture and latching mechanisms.	C–D
	1213: Mechanical validation and functional testing.	D
WP-1220 Fluid Transfer	**1221:** Refueling pumps and transfer valves.	A–C
	1222: Transfer storage and reception tanks.	B–D
	1223: Flow control system and leak testing.	D
WP-1230 Prox. Sensors	**1231:** LiDAR and optical camera integration.	B–C
	1232: Short-range radar systems definition.	B–C
	1233: Data Processing Unit development.	C–D

Continued on next page

WP	Scope	Phases
WP-1310 Integ. & Test	**1311:** Subsystem-level AIT execution.	C–D
	1312: Full spacecraft system integration.	D
	1313: Environmental system testing (Vibe, EMC).	D
WP-1320 Nominal Orbit	**1321:** LEOP operations.	E
	1322: Spacecraft commissioning.	E
	1323: Routine operations and health monitoring.	E
WP-1330 Rendezvous	**1331:** Orbit phasing and far-approach manoeuvres.	E
	1332: Close proximity navigation.	E
	1333: Mating and capture execution.	E
WP-1340 Refueling Ops	**1341:** Mechanical connection and seal checks.	E
	1342: Active fluid transfer and telemetry monitoring.	E
	1343: Demating and separation operations.	E
WP-1350 Disposal Plan	**1351:** Spacecraft passivation procedures.	F
	1352: Graveyard orbit injection manoeuvres.	B, F
	1353: EOM declaration.	F
WP-2000 Ground Segment		
WP-2100 GS Dev.	**2110:** Ground Segment requirements and interfaces.	C
	2120: Procurement of stations and equipment.	C–D
	2130: Ground Segment validation and end-to-end tests.	D
WP-2200 Mission Ctrl	**2210:** Satellite control software (TM/TC).	C–E
	2220: Refueling specific telemetry dashboards.	C–E
	2230: Operational procedures and simulations.	C–D
WP-3000 Launcher		
WP-3100 Selection	**3110:** Identify launcher candidates.	A–B
	3120: Trade-off and mission compatibility.	A–B
	3130: Launch service contract preparation.	B–C
WP-3200 Launch Camp.	**3210:** Shipping and handling to the launch site.	D
	3220: Launch site operations and integration.	D
	3230: Propellant loading and final checks.	D
WP-3300 Launch Ops	**3310:** Countdown and ascent phase.	D–E
	3320: Trajectory monitoring and flight tracking.	E
	3330: 3rd stage operations and spacecraft separation.	E

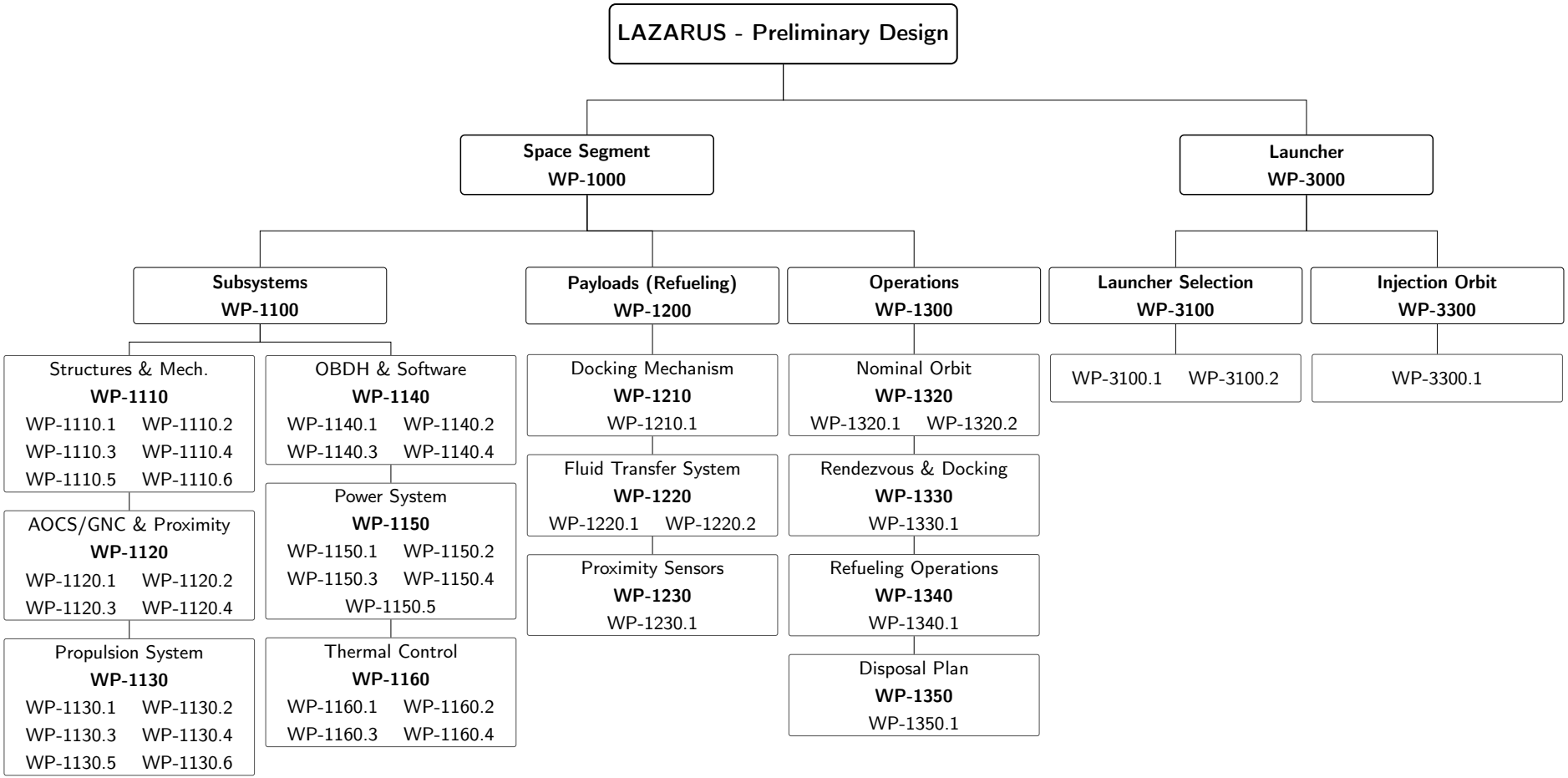


Figure 7.1: WBS for the LAZARUS Preliminary Design.

7.2 WBS of the complete Project

The LAZARUS WBS covers the entire mission lifecycle (Phases 0+A to F), establishing the baseline for planning, cost control, and risk management.

Structure and Numbering. The WBS uses a four-digit hierarchical code (thousands for Level 1, hundreds for Level 2, tens for Level 3, and units for Level 4):

- **Level 1:** Management (WP-0000), Space Segment (WP-1000), Ground Segment (WP-2000), and Launcher (WP-3000).
- **Levels 2 & 3:** Functional breakdowns (e.g., WP-1100 Platform, WP-1120 AOCS/GNC & Proximity).
- **Level 4:** Specific actionable packages (e.g., WP-1122 for GNC algorithm design).

Upper levels (1–3) are shown in Figure 7.2, branch details in Figures 7.3–7.9, and package scopes with ECSS phases in Table 7.2.

Assumptions.

- The project scope extends through EOL disposal (Phase F, WP-1350).
- Subsystem equipment, whether developed in-house or procured, is integrated into its respective subsystem work package.
- Launcher selection, campaign, and operations are consolidated entirely within the Launcher segment (WP-3000).

Table 7.2: Detailed scope of the LAZARUS WBS down to Level 4, including specific ECSS phases. Launch operations have been shifted to WP-3000.

WP	Scope	Phases
WP-0000 Management		
WP-0100 Planning	**0110:** Prepare the Project Management Plan.	0+A–B
	0120: Define the project phasing and milestones.	0+A–B
	0130: Develop and maintain the Project Work Plan.	0+A–F
	0140: Elaborate the PBS and WBS.	0+A–B
WP-0200 Risk Mgmt	**0210:** Identify main technical and programmatic risks.	0+A–B
	0220: Assess impact and probability (Risk Matrix).	0+A–F
	0230: Define mitigation actions and assign owners.	0+A–F
	0240: Maintain risk register and follow up.	0+A–F
WP-0300 Cost & Sched.	**0310:** Prepare Manpower and Cost Plan.	0+A–B
	0320: Monitor manpower and schedule vs baseline.	0+A–F
	0330: Forecast cost at completion and report.	0+A–F
WP-0400 Reviews	**0410:** Manage documents and configuration.	0+A–F
	0420: Prepare and hold Mission Definition Review (MDR).	0+A
	0430: Prepare and hold PRR.	A
	0440: Prepare and hold SRR.	B
	0450: Prepare and hold PDR.	B
	0460: Prepare and hold CDR.	C
	0470: Prepare and hold QR.	D
	0480: Prepare and hold FAR.	D
WP-1000 Space Segment		
WP-1110 Structures	**1111:** Define requirements and trade-offs.	A–B
	1112: Detailed design and structural analysis, Finite Element Method (FEM).	B–C
	1113: Manufacturing of structural components.	C–D
	1114: Assembly, Integration and Testing (AIT).	D
WP-1120 AOCS/GNC & Prox	**1121:** Selection of sensors and actuators.	A–B
	1122: Design of GNC algorithms.	B–C
	1123: Development of the proximity vision system.	C–D
	1124: Hardware-in-the-Loop (HWIL) testing.	D

Continued on next page

WP	Scope	Phases
WP-1130 Propulsion	**1131:** Thrusters and propellant tanks sizing.	A–C
	1132: Feed system and valves detailed design.	B–C
	1133: Pressurization system integration.	C–D
	1134: Subsystem firing and qualification tests.	D
WP-1140 OBDH & SW	**1141:** On-Board Computer architecture and selection.	A–C
	1142: Data buses definition (CAN, SpaceWire).	B–C
	1143: Flight Software (FSW) development.	B–D
	1144: Software validation and V&V testing.	D
WP-1150 Power	**1151:** Solar arrays sizing and deployment mech.	A–C
	1152: Battery sizing and PCDU design.	B–C
	1153: Power distribution network layout.	C–D
	1154: Subsystem electrical integration.	D
WP-1160 Thermal	**1161:** Definition of Multi-layer insulation (MLI) and optical coatings.	B–C
	1162: Sizing of heaters and thermistors.	B–C
	1163: Thermal node analysis (Hot/Cold cases).	B–C
	1164: Thermal Vacuum test (TVAC).	D
WP-1210 Docking Mech	**1211:** Mechanical design and interfaces.	A–C
	1212: Actuators, capture and latching mechanisms.	C–D
	1213: Mechanical validation and functional testing.	D
WP-1220 Fluid Transfer	**1221:** Refueling pumps and transfer valves.	A–C
	1222: Transfer storage and reception tanks.	B–D
	1223: Flow control system and leak testing.	D
WP-1230 Prox. Sensors	**1231:** LiDAR and optical camera integration.	B–C
	1232: Short-range radar systems definition.	B–C
	1233: Data Processing Unit development.	C–D

Continued on next page

WP	Scope	Phases
WP-1310 Integ. & Test	**1311:** Subsystem-level AIT execution.	C–D
	1312: Full spacecraft system integration.	D
	1313: Environmental system testing (Vibe, EMC).	D
WP-1320 Nominal Orbit	**1321:** Launch and Early Orbit Phase (LEOP) operations.	E
	1322: Spacecraft commissioning.	E
	1323: Routine operations and health monitoring.	E
WP-1330 Rendezvous	**1331:** Orbit phasing and far-approach manoeuvres.	E
	1332: Close proximity navigation.	E
	1333: Mating and capture execution.	E
WP-1340 Refueling Ops	**1341:** Mechanical connection and seal checks.	E
	1342: Active fluid transfer and telemetry monitoring.	E
	1343: Demating and separation operations.	E
WP-1350 Disposal Plan	**1351:** Spacecraft passivation procedures.	F
	1352: Graveyard orbit injection manoeuvres.	B, F
	1353: End of Mission (EOM) declaration.	F
WP-2000 Ground Segment		
WP-2100 GS Dev.	**2110:** Ground Segment requirements and interfaces.	C
	2120: Procurement of stations and equipment.	C–D
	2130: Ground Segment validation and end-to-end tests.	D
WP-2200 Mission Ctrl	**2210:** Satellite control software (TM/TC).	C–E
	2220: Refueling specific telemetry dashboards.	C–E
	2230: Operational procedures and simulations.	C–D
WP-3000 Launcher		
WP-3100 Selection	**3110:** Identify launcher candidates.	A–B
	3120: Trade-off and mission compatibility.	A–B
	3130: Launch service contract preparation.	B–C
WP-3200 Launch Camp.	**3210:** Shipping and handling to the launch site.	D
	3220: Launch site operations and integration.	D
	3230: Propellant loading and final checks.	D
WP-3300 Launch Ops	**3310:** Countdown and ascent phase.	D–E
	3320: Trajectory monitoring and flight tracking.	E
	3330: 3rd stage operations and spacecraft separation.	E

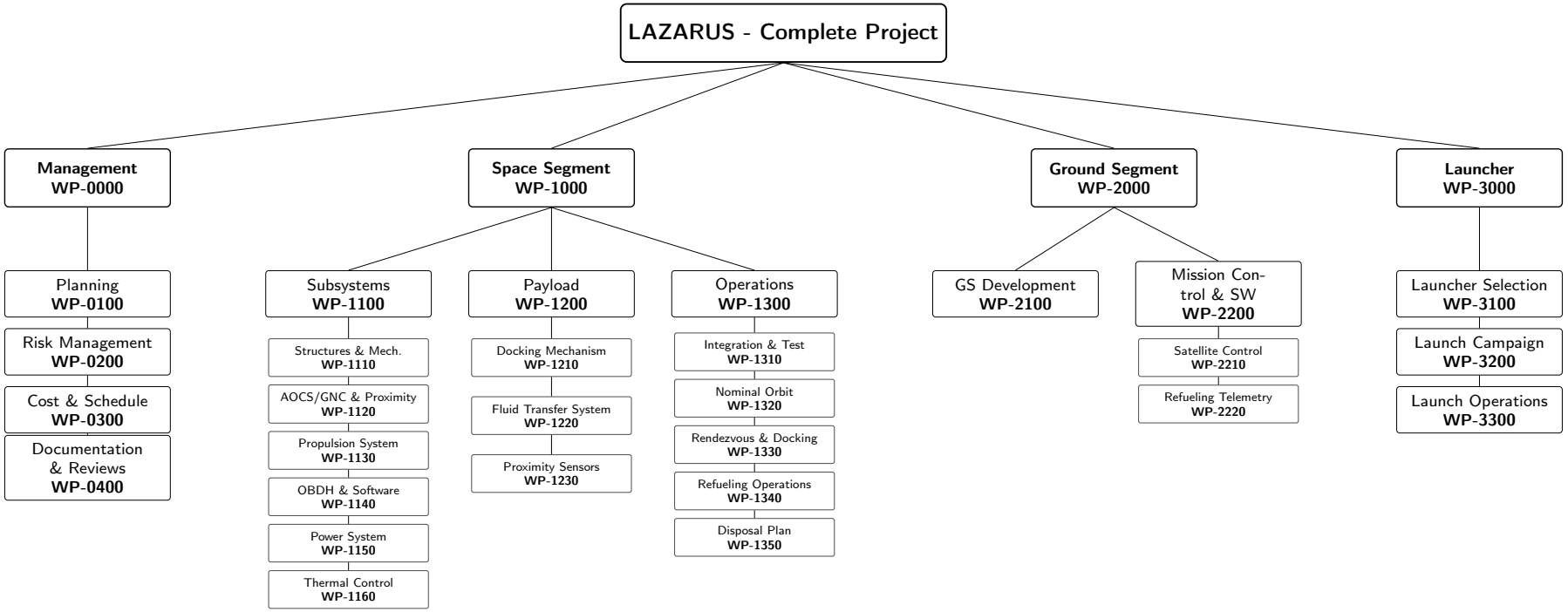


Figure 7.2: WBS for the LAZARUS Mission.

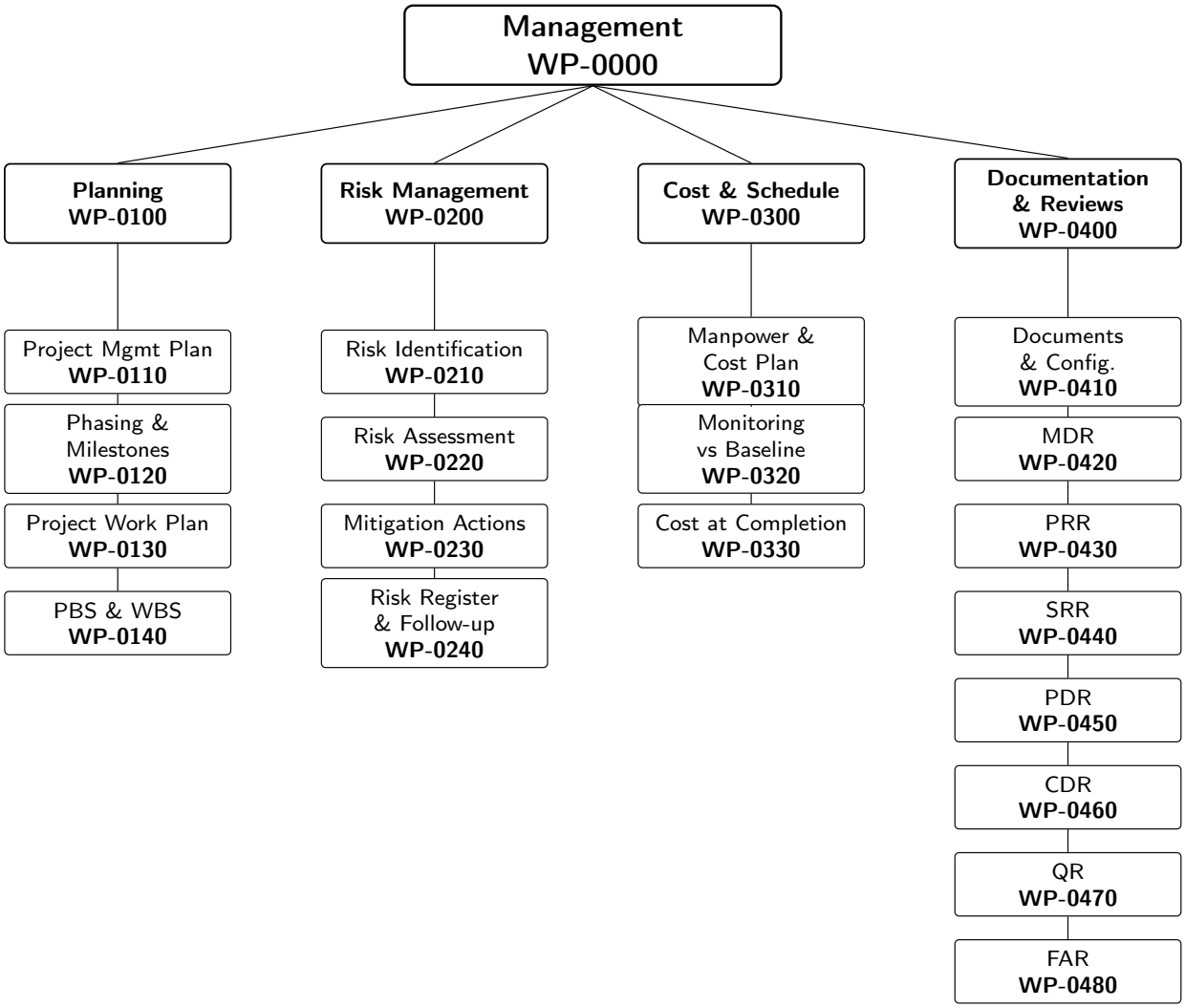


Figure 7.3: WBS of the Management branch (WP-0000).

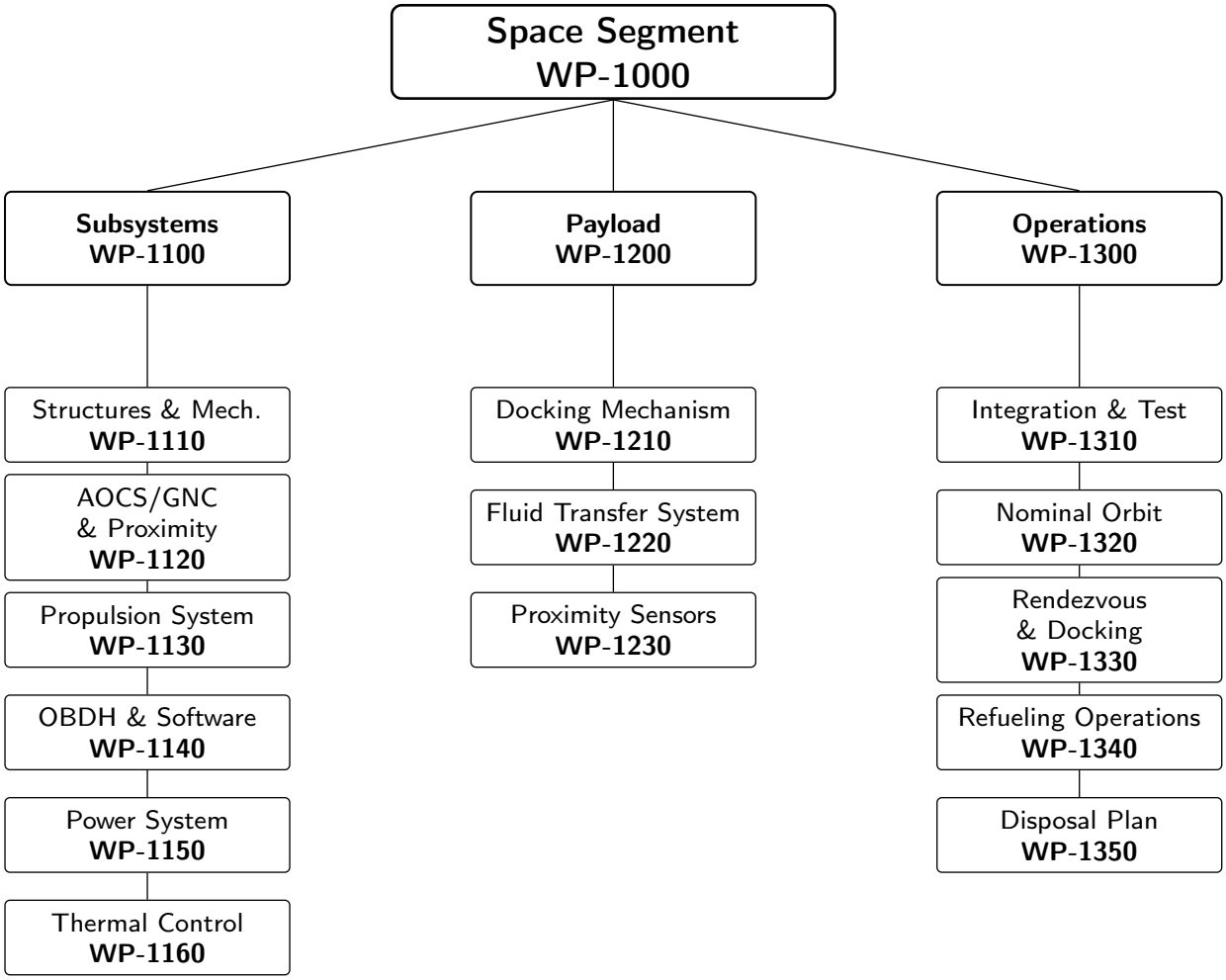


Figure 7.4: WBS of the Space Segment (WP-1000).

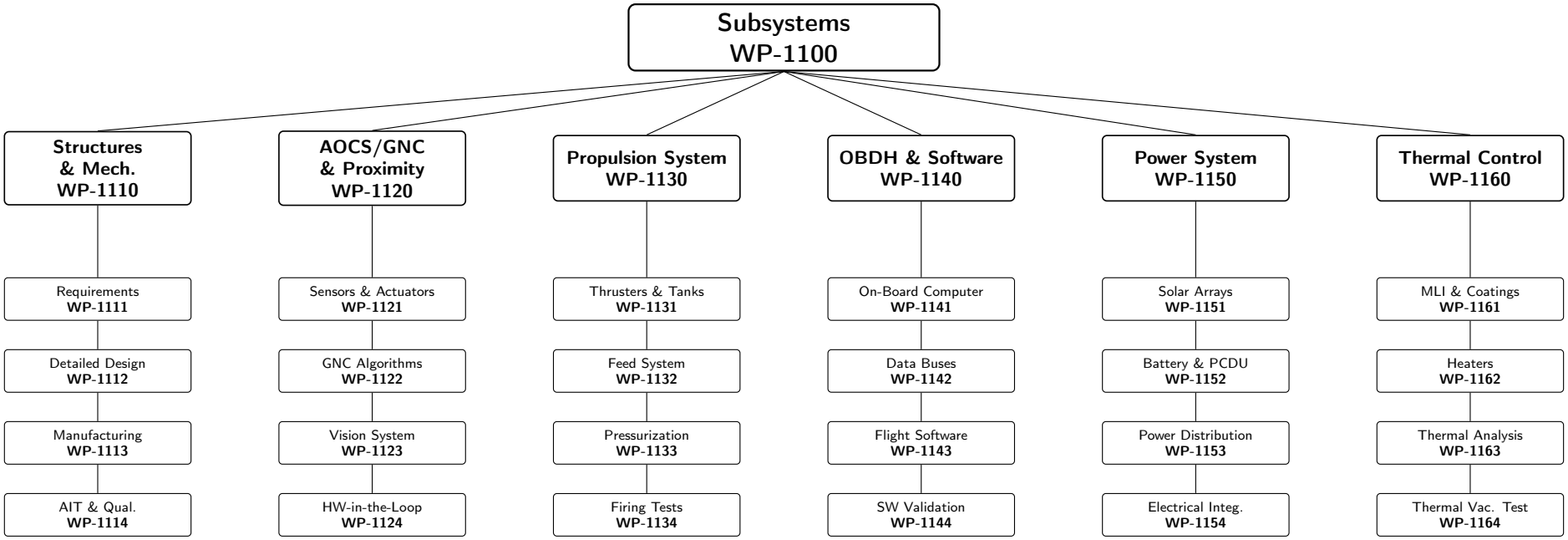


Figure 7.5: WBS of the Subsystems (WP-1100), down to Level 4.

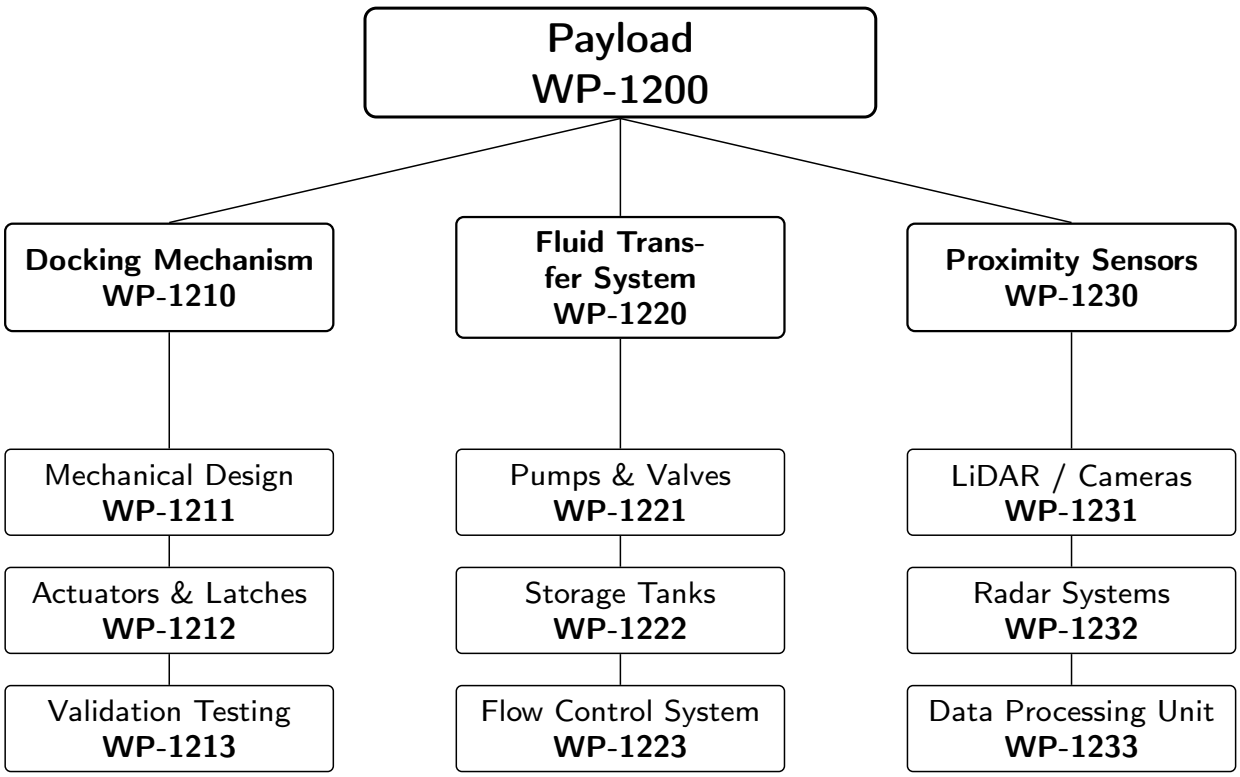


Figure 7.6: WBS of the Payload (WP-1200), down to Level 4.

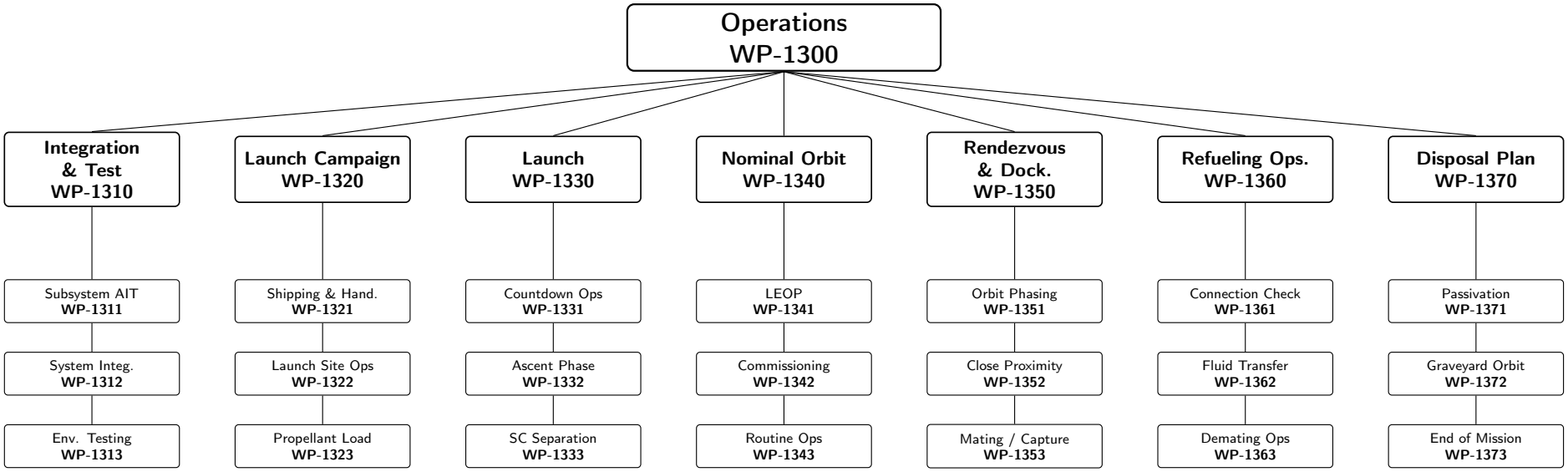


Figure 7.7: WBS of the Operations (WP-1300), down to Level 4.

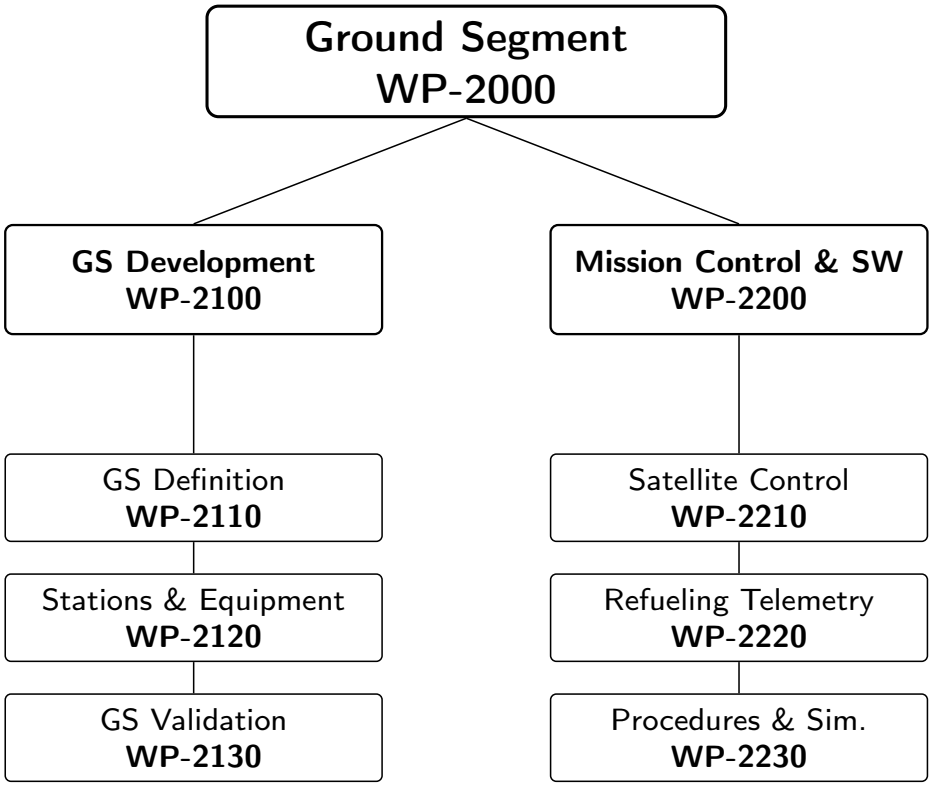


Figure 7.8: WBS of the Ground Segment (WP-2000).

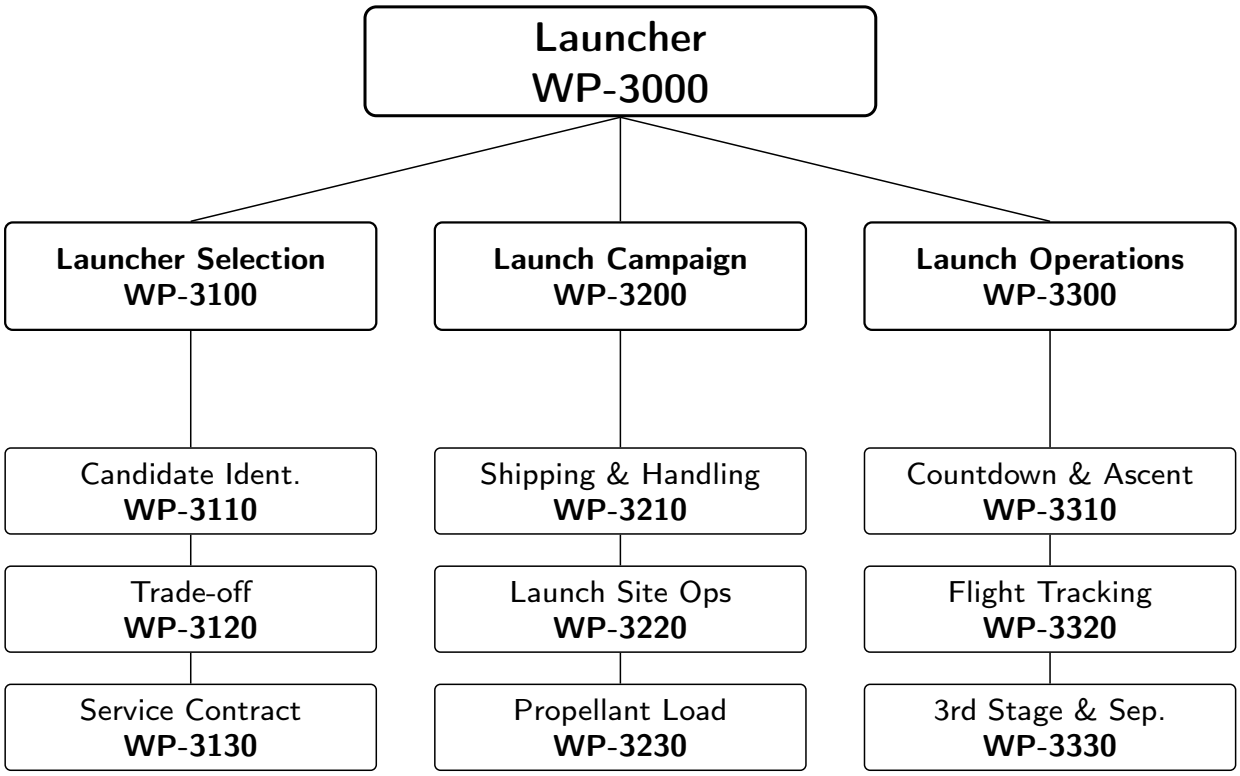


Figure 7.9: WBS of the Launcher (WP-3000).

References

[1] Bombardelli C. Astrodynamics and Attitude Dynamics: Gravity Assist and Flyby Dynamics. Madrid, Spain; 2023. Lecture notes, Master's Degree in Aeronautical Engineering.

[2] Bombardelli C. Astrodynamics and Attitude Dynamics: The Two-Body Problem (Review). Madrid, Spain; 2025. Lecture notes, Master's Degree in Aeronautical Engineering.

[3] del Cura Velayos JM. Systems Engineering and Preliminary Design of Space Vehicles. Topic 3: Preliminary Design of a Space Mission. Presentation. Madrid, Spain; 2020. Lecture notes.

[4] Hardware Upgrade. ESA e ClearSpace svilupperanno un satellite per prolungare la vita utile di altri satelliti in orbita geostazionaria; 2024. Accessed: 2026-10-06. Online news article. Available from: https://www.hwupgrade.it/news/scienza-tecnologia/esa-e-clearspace-svilupperanno-un-satellite-per-prolungare-la-vita-utile-di-altri-satelliti-in-orbita-geostazionaria_159822.html.

[5] Hardware Upgrade. Un razzo spaziale Falcon 9 ha lanciato la missione Northrop Grumman MRV-MEP per prolungare la vita dei satelliti; 2024. Accessed: 2026-10-06. Online news article. Available from: https://www.hwupgrade.it/news/scienza-tecnologia/un-razzo-spaziale-falcon-9-ha-lanciato-la-missione-northrop-grumman-mrv-mep-per-prolungare-la-vita-dei-satelliti_156697.html.

[6] Skybrokers. GEO Satellites Database and Information; 2026. Accessed: 2026-10-06. Web page. Available from: https://sky-brokers.com/geo-satellites/.

[7] Zorita D, Agenjo A, Cazorla C, Llorente S, Del Cura JM, O'Dwyer A, et al. Autonomous Orbit Control Strategy for Accurate Large Delta-V Manoeuvres in Planck Spinning S/C. In: IFAC Automatic Control in Aerospace. Saint-Petersburg, Russia; 2004. .

[8] Miravet C, Pascual L, Krouch E, Del Cura JM. An Image-Based Sensor System for Autonomous Rendez-Vous with Uncooperative Satellites. In: 7th International ESA Conference on Guidance, Navigation & Control Systems (GNC 2008). Tralee, County Kerry, Ireland; 2008. .

[9] Lozano González JM, Praile C, Erb S, Del Cura JM, Rodriguez G, Weikert S. Optimisation of NS, EW Station-Keeping Manoeuvres for GEO Satellites Using Electric Propulsion (OPASKEP). ESA; 2006.

[10] Tarabini L, Gil J, Gandia F, Molina MÁ, Del Cura JM, Ortega G. Ground guided CX-OLEV rendez-vous with uncooperative geostationary satellite. Acta Astronautica. 2007;61(4):312-25.

[11] Jehn R, Agapov V, Hernández C. End-of-Life Disposal of Geostationary Satellites. In: Proceedings of the Fourth European Conference on Space Debris. ESA SP-587. Darmstadt, Germany; 2005. .

[12] Del Cura JM, Saavedra G, Sánchez-Maestro R, Sebastián A, Tarabini L, Ortega G, et al. ConeXpress Orbital Life Extension Vehicle (CX-OLEV) GNC. In: 57th International Astronautical Congress. IAC-06-C1.8.08. Valencia, Spain; 2006. .